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A simple matching algorithm for regular bipartite graphs[☆]

Kazuhisa Makino, Takashi Takabatake^{*}, Satoru Fujishige

Division of Systems Science, Graduate School of Engineering Science, Osaka University, Toyonaka, Osaka 560-8531, Japan

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Abstract

We consider the perfect matching problem for a Δ -regular bipartite graph with n vertices and m edges, i.e., $\frac{1}{2}n\Delta = m$, and present a new $O(m + n \log n \log \Delta)$ algorithm. Cole and Rizzi, respectively, gave algorithms of the same complexity as ours, Schrijver also devised an $O(m\Delta)$ algorithm, and the best existing algorithm is Cole, Ost, and Schirra's $O(m)$ algorithm. Extending Gabow's perfect matching algorithm for 2^t -regular bipartite graph with a positive integer t and using Cole and Hopcroft's edge-sparsification technique, we show another approach to the perfect matching problem, which results in a simple algorithm that employs no sophisticated data structure such as dynamic tree and splay tree.

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1. Introduction

It is well known that any regular bipartite graph has a perfect matching and that the perfect matching problem for regular bipartite graphs is related to the edge-coloring problem for general bipartite graphs. For example, Kapoor and Rizzi [6] showed that the edge-coloring problem for a bipartite graph G with m edges and maximum degree Δ can be solved in $T + O(m \log \Delta)$ time, where T is the time required to

compute a perfect matching in a k -regular bipartite graph with $O(m)$ edges and $k \leq \Delta$.

In this paper, we consider the perfect matching problem for regular bipartite graphs with possible multiple edges. The perfect matching problem for regular bipartite graphs has been studied and a lot of algorithms have been proposed in the literature (see, e.g., [1–5,7,8]). Cole and Hopcroft [2] presented an $O(m + n \log n \log^2 \Delta)$ algorithm for computing a perfect matching in a Δ -regular bipartite graph with n vertices and m edges. This time complexity was improved by Cole [1] and Rizzi [7] to $O(m + n \log n \log \Delta)$. Schrijver [8] also presented an $O(m\Delta)$ algorithm and Cole, Ost, and Schirra [3] obtained an $O(m)$ algorithm by improving Schrijver's algorithm [8] by using splay trees, one for each chain, as a data structure.

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^{*} Corresponding author.

E-mail addresses: makino@sys.es.osaka-u.ac.jp (K. Makino), takabatake@sys.es.osaka-u.ac.jp (T. Takabatake), fujishige@sys.es.osaka-u.ac.jp (S. Fujishige).

Extending Gabow's algorithm [5] for computing a perfect matching in a 2^t -regular bipartite graph for a positive integer t and using Cole and Hopcroft's edge-sparsification technique [2], we show another approach to the perfect matching problem, which results in a new simple $O(m + n \log n \log \Delta)$ algorithm. It uses no sophisticated data structure such as dynamic tree and splay tree employed in [1,3], and runs in linear time for regular bipartite graphs with $\Delta \geq \log n \log \Delta$. It is expected that our algorithm will achieve good practical performance, while efficient implementations and computational experiments of the above algorithms (including ours) deserve further study.

The rest of the paper is organized as follows. In Section 2 we first present an $O(m \log n)$ algorithm for computing a perfect matching in a regular bipartite graph, which will give us a basis for a faster algorithm. By using the edge-sparsification technique due to Cole and Hopcroft [2], we present a faster algorithm for computing a perfect matching in a regular bipartite graph in Section 3.

2. An $O(m \log n)$ algorithm

In this section, we present an $O(m \log n)$ algorithm for computing a perfect matching in a regular bipartite graph, which will be made faster in the next section. Let $G = (V, E)$ be a Δ -regular bipartite graph with n vertices and m edges. Here V has a bipartition (V^+, V^-) , i.e., any edge $e \in E$ is incident to a vertex in V^+ and a vertex in V^- . Note that $m = n\Delta/2$ and $|V^+| = |V^-| = n/2$.

Let us first note that a perfect matching in a 2^t -regular bipartite graph G with a positive integer t can be computed in linear time [5]. We first find a Eulerian orientation of G that consists of Eulerian tours, one for each connected component of G , and then remove those edges in G that are oriented from V^- to V^+ . This gives a 2^{t-1} -regular subgraph of G . By repeating this procedure t times, we finally obtain a 1-regular subgraph (i.e., a perfect matching) of G . Since a Eulerian orientation of G can be found in $O(m)$ time, Gabow's algorithm [5] requires $O(m + \frac{1}{2}m + \frac{1}{4}m + \dots) = O(m)$ time. However, if a given regular graph is not 2^t -regular for any positive integer t , the above algorithm does not work, since it ends

up with a $(2k + 1)$ -regular subgraph for some integer $k \geq 1$ that has no Eulerian orientation.

Therefore, our algorithm first makes G 2^t -regular by adding new edges to G . More precisely, let $G = (V, E)$ be a Δ -regular bipartite graph such that $2^{t-1} < \Delta \leq 2^t$ for some positive integer t . Let M_1 be a perfect matching of the complete bipartite graph $K_{n/2, n/2}$ with the bipartition (V^+, V^-) of V . We first construct a 2^t -regular bipartite graph $\widehat{G} = (V, \widehat{E})$ by adding $2^t - \Delta$ copies of M_1 to G . A new edge $\widehat{e} \in \widehat{E} \setminus E$ is called *dummy* if there exists no edge $e \in E$ such that e and $\widehat{e}$ are parallel. Note that the number of dummy edges in $\widehat{G}$ is at most $\frac{1}{2}n(2^t - \Delta) < |\widehat{E}|/2$. We then apply a variant of Gabow's algorithm [5] to $\widehat{G}$ to find a perfect matching M_2 in $\widehat{G}$. In this variant, given a Eulerian orientation, if the number of dummy edges oriented from V^+ to V^- is at most the half of the number of all dummy edges, to get M_2 we remove those edges that are oriented from V^- to V^+ ; otherwise we remove those edges that are oriented from V^+ to V^- . Note that M_2 has at most $\frac{1}{4}n (= |V^+|/2)$ dummy edges. In other words, the size of the matching formed by the non-dummy edges in M_2 is at least $|V^+| - |V^+|/2 (= |V^+|/2)$. We again construct a 2^t -regular bipartite graph $\widehat{G}$ by adding $2^t - \Delta$ copies of M_2 to G , and apply Gabow's algorithm to it. Let M_3 be the obtained matching in $\widehat{G}$. Since $\widehat{G}$ contains at most $\frac{1}{4}n(2^t - \Delta) < |\widehat{E}|/4$ dummy edges, M_3 has at most $|V^+|/4$ dummy edges (i.e., the size of the matching formed by non-dummy edges in M_3 is at least $|V^+| - |V^+|/4 (= 3|V^+|/4)$). By repeating this procedure at most $\lceil \log |V^+| \rceil$ times, we finally obtain a perfect matching of G .

Formally, the algorithm described above can be given as in Algorithm 1.

Note that when M in Step 4 contains no dummy edge before we get $k > \lceil \log |V^+| \rceil$, we can output M and halt.

Theorem 2.1. *Algorithm ADD-SPLIT correctly computes a perfect matching of G in $O(m \log n)$ time.*

Proof. Since the above argument shows the correctness of the algorithm, we consider its time complexity. It is clear that Steps 1, 2 and 4 require $O(m)$ time. From the result in [5], Step 3 can be done in $O(m)$ time. Since the number of iterations between

Algorithm ADD-SPLIT

Input : A Δ -regular bipartite graph $G = (V, E)$ such that $2^{t-1} < \Delta \leq 2^t$ for some positive integer t .

Output : A perfect matching M in G .

Step 1 : Compute a perfect matching M of $K_{n/2, n/2}$ and put $k := 1$.

Step 2 : Construct a 2^t -regular bipartite graph $\widehat{G} = (V, \widehat{E})$ by adding $2^t - \Delta$ copies of M to G .

Step 3 : While $\widehat{G}$ is not 1-regular do

(3-I) Find a Eulerian orientation of $\widehat{G}$.

(3-II) If the number of dummy edges oriented from V^+ to V^- is at most the half of the number of all dummy edges in $\widehat{G}$, then remove those edges that are oriented from V^- to V^+ ; otherwise remove those edges that are oriented from V^+ to V^- . Denote the resultant graph by $\widehat{G} = (V, \widehat{E})$ again.

Step 4 : Put $M := \widehat{E}$ and $k := k + 1$. If $k \leq \lceil \log |V^+| \rceil$, then go to Step 2. Otherwise return M and halt.

Algorithm 1.

Step 2 and Step 4 is $\lceil \log n \rceil$, the algorithm requires $O(m \log n)$ time in total. $\square$

3. An $O(m + n \log n \log \Delta)$ algorithm

An edge-sparsification technique for regular bipartite graphs was proposed by Cole and Hopcroft [2] and has been used to obtain faster algorithms for computing a perfect matching in a regular bipartite graph [1–3,7]. We also employ this technique to devise a faster version of our algorithm.

Given a Δ -regular bipartite graph $G = (V, E)$ with $2^{t-1} < \Delta \leq 2^t$ for some positive integer t , the *edge-sparsification* produces a subgraph $G^* = (V, E^*)$ of G having an edge capacity function $c: E^* \rightarrow \{1, 2, 2^2, \dots, 2^t\}$ such that

- (i) for each $v \in V$, the sum of the edge capacities $c(e)$ for all edges e incident to v is equal to Δ , i.e.,

$$\sum \{c(e) \mid e \in E^* \text{ is incident to } v\} = \Delta,$$

- (ii) for each $\ell \in \{0, 1, \dots, t\}$, the set of all edges e with $c(e) = 2^\ell$ forms a forest (i.e., it contains no cycle).

Cole and Hopcroft [2] showed that the edge-sparsification can be done in linear time without using any sophisticated data structure. We identify an edge e having capacity $c(e)$ with parallel edges formed by $c(e)$ copies of e . It follows from (i) that G^* can be regarded as a Δ -regular graph, and hence it always contains a perfect matching which can also

be regarded as a perfect matching in the original graph G . By (ii), G^* has at most $(n-1)\lceil \log \Delta \rceil$ edges.

Intuitively speaking, we apply Algorithm ADD-SPLIT to graph G^* instead of G . Since G^* has at most $(n-1)\lceil \log \Delta \rceil$ edges, the algorithm requires $O(m + (n-1)\log \Delta \times \log n) = O(m + n \log n \log \Delta)$ time, where the time required for the edge-sparsification is $O(m)$.

Let us assume that $2^{t-1} < \Delta < 2^t$ for some positive integer t , since a perfect matching of a 2^t -regular bipartite graph can be obtained in linear time [5]. We need some notations to describe the details of our faster algorithm. Apply the edge-sparsification procedure to a graph formed by $(2^t - \Delta) (\leq 2^{t-1})$ copies of a perfect matching M in $K_{n/2, n/2}$. Denote by M^* the resultant graph with edge capacities. Let $\widehat{G} = (V, \widehat{E})$ be the graph obtained by adding M^* to G^* , and let $\widehat{E}_\ell$ ($\ell = 0, 1, \dots, t-1$) be the set of all edges e in $\widehat{E}$ with $c(e) = 2^\ell$. Note that $\widehat{G}$ with edge capacities can be regarded as a 2^t -regular graph. A 2^{t-1} -regular subgraph $H = (V, F)$ of $\widehat{G}$ can be computed as follows.

- (1) Compute a Eulerian orientation of $\widehat{E}_0$ and choose those edges that are oriented from V^+ to V^- (or V^- to V^+). Denote by R_1 the set of such edges.
- (2) For each $\ell \in \{0, 1, \dots, t-2\}$ define

$$F_\ell = \begin{cases} R_1 \cup \widehat{E}_1 & \text{if } \ell = 0, \\ \widehat{E}_{\ell+1} & \text{otherwise,} \end{cases} \quad (3.1)$$

and put $F = \bigcup_{\ell=0}^{t-2} F_\ell$.

Algorithm BITWISE-ADD-SPLIT

Input : A Δ -regular bipartite graph $G = (V, E)$ such that $2^{t-1} < \Delta \leq 2^t$ for some positive integer t .

Output : A perfect matching M in G .

Step 0 : If $\Delta = 2^t$, then compute a perfect matching M of G by applying Gabow's algorithm to G , and halt.

Step 1 : Apply the edge-sparsification procedure to G . Denote by G^* the resultant graph with edge capacities. Let M be a perfect matching of $K_{n/2, n/2}$ and put $k := 1$.

Step 2 : Let M^* be the graph, with edge capacities, obtained from $2^t - \Delta$ copies of M by the edge-sparsification procedure and construct a 2^t -regular bipartite graph $\widehat{G} = (V, \widehat{E})$ by adding M^* to G^* .

Step 3 : Put $R_0 := \emptyset$. For $\ell = 0, 1, \dots, t-1$ do

(3-I) Find a Eulerian orientation of $\widehat{E}_\ell \cup R_\ell$.

(3-II) If the number of dummy edges oriented from V^+ to V^- is at most the half of the number of all dummy edges, then remove those edges that are oriented from V^- to V^+ ; otherwise remove those edges that are oriented from V^+ to V^- . Denote the resultant edge set by $R_{\ell+1}$.

Step 4 : Put $M := R_t$ and $k := k + 1$. If $k \leq \lceil \log |V^+| \rceil$, then go to Step 2. Otherwise return M and halt.

Algorithm 2.

Note that each edge $e \in F_\ell$ ($\ell = 0, 1, \dots, t-2$) has the capacity $c(e) = 2^\ell$. Therefore, Step 3 in Algorithm ADD-SPLIT can be performed as follows. Let $R_0 = \emptyset$ and for each $\ell = 1, 2, \dots, t-1$ let R_ℓ be the edge set obtained from a Eulerian orientation of $\widehat{E}_{\ell-1} \cup R_{\ell-1}$. Then we get a perfect matching of $\widehat{G}$ from a Eulerian orientation of $\widehat{E}_{t-1} \cup R_{t-1}$.

Now, we have Algorithm 2. Similarly as in Algorithm ADD-SPLIT, if M in Step 4 contains no dummy edge before we get $k > \lceil \log |V^+| \rceil$, we can output M and halt. Note further that in Step 2, the edge-sparsification for $2^t - \Delta$ copies of M can be done in $O(n \log \Delta)$ time by considering the binary representation of $2^t - \Delta$.

Let us examine the properties of $\widehat{E}_\ell \cup R_\ell$ ($\ell = 0, 1, \dots, t-1$) in Step 3 of the algorithm.

Lemma 3.1. Define $H_\ell = (V, \widehat{E}_\ell \cup R_\ell)$ for $\ell = 0, 1, \dots, t-1$. Then the following two statements hold for $\ell = 0, 1, \dots, t-1$:

- (i) The degree of each vertex in H_ℓ is even.
- (ii) H_ℓ has at most $3n$ edges.

Proof. Since $\widehat{G}$ is 2^t -regular, the first statement holds. The second one is shown by induction on ℓ , since $\widehat{E}_\ell$ and R_ℓ have at most $(n-1) + n/2 \leq 3n/2$ and $3n/2$ edges, respectively. $\square$

Theorem 3.2. Algorithm BITWISE-ADD-SPLIT finds a perfect matching of a Δ -regular bipartite graph G in $O(m + n \log n \log \Delta)$ time.

Proof. Since the discussion in Sections 2 and 3 shows the correctness of the algorithm, we only consider its time complexity. Steps 0 and 1 can be done in $O(m)$ time [5,2], and Step 2 requires $O(n \log \Delta)$ time, since $|E^*|, |M^*| \leq (n-1) \log \Delta$. It follows from Lemma 3.1 that Step 3 requires $O(n \log \Delta)$ time. Moreover, Step 4 requires $O(n)$ time. Since the number of iterations between Step 2 and Step 4 is $\lceil \log n \rceil$, the algorithm requires $O(m + n \log n \log \Delta)$ time in total. $\square$

By combining it with the result by Kapoor and Rizzi [6], we have the following corollary.

Corollary 3.3. A minimum edge-coloring of a Δ -bipartite graph can be found in $O((m + n \log n) \log \Delta)$ time.

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